Patent Portfolio Roadmap

Source synthesis, opportunity ranking, and next-system recommendation

Prepared 19 October 2026
Source basis: four user-provided patent development packages

Decision-support package for selecting the next system or patent concept to develop.

STATUS | Portfolio synthesis and concept hypotheses only
No novelty, freedom-to-operate, engineering-performance, or filing-readiness conclusion is implied.

1. Executive decision

Recommended next development: a platform-neutral, non-operational verification system tentatively named TRACEGATE - a Configuration-Linked Autonomy Verification and Change-Impact Gate. Its role would be to decide which safety and reliability evidence must be re-run when a robot's hardware, sensors, payload, software, model, calibration, or approved operating assumptions change. It would produce a traceable readiness package for a defined configuration and use case, rather than plan routes or command a vehicle.

Why this is the strongest next step The four packages already cover vehicle-level autonomy, swarm allocation, aircraft flight management, and vehicle configuration/crew-safety checks. Their shared gap is a reusable way to connect a change in a real configuration to the exact safety claims, assumptions, and tests that must be revalidated before that configuration is treated as ready.

This is a development hypothesis, not a conclusion that TRACEGATE is new or patentable. A first-pass public patent screen found adjacent work on robot verification/authentication, robotic assurance models, and autonomous-vehicle digital twins. A claim-focused prior-art search must therefore target the specific dependency propagation and evidence-selection mechanism, not broad claims to robot safety, readiness, configuration checking, or digital twins.

Decision in one sentence

Build a small proof-of-concept that maps configuration changes to affected safety requirements and targeted repeatable tests, then commission a patentability search and claim chart before investing in a full application.

2. What the four source packages currently cover

Source package	Current concept in the supplied files	Reusable technical assets	Main open issue
AEROSYNC - Quadrotor swarm	Cooperative quadrotors for authorized civilian survey, inspection, mapping, environmental monitoring, and emergency assessment. Shared mission state; capability/energy/uncertainty-aware task allocation; independent safety supervision; bounded contingencies.	Mission-state schema; uncertainty-aware task assignment; fault and communications-loss test cases; fleet logs; safety-gate pattern.	Concept-stage. Cooperative task allocation is known prior art; the specific combination needs a claim chart and measured evidence.
AQUAVANTIS-G	The actual package text describes a civilian, non-weaponized unmanned ground vehicle with traversability, localization confidence, energy, payload constraints, recovery feasibility, and a safety supervisor.	Configuration/health inputs; hard safety constraints; unknown-state handling; recovery planning; audit logs; engineering verification matrix.	Identity mismatch: the package filename and earlier topic may suggest an underwater vehicle, but the supplied README/specification say UGV. Confirm intended identity.
SAGE-APC	Condition-aware modular protection and crew-safety management. A versioned configuration-and-condition ledger links module identity, zone, installation evidence, condition, approved variant,	Configuration ledger; statuses VERIFIED / RESTRICTED / UNKNOWN / INSPECTION REQUIRED; fault-aware evidence handling; service and inspection records.	A preliminary memo already recognizes adjacent modular vehicle, armor-damage sensing, and vehicle damage detection references. Do not claim modular armor alone.

Source package	Current concept in the supplied files	Reusable technical assets	Main open issue
	and crew-safety constraints.		
UCAS - revised 2026-10-06	Uncrewed aircraft autonomy with sensor provenance and environment uncertainty, AI candidate planning, deterministic safety-envelope gate, health-aware constraint updates, authenticated mission contracts, communications degradation, state reconciliation, and evidence logging.	Requirement-to-evidence links; confidence/provenance fields; model lifecycle; change/reconciliation events; shadow evaluation and regression tests.	Very broad, technically overlapping with known autonomy, mission planning, sensor-fusion, and health-aware contingency work. Novelty remains unverified.

3. Cross-portfolio architecture map

Across the source documents, the common architectural motif is: observe and estimate state; retain confidence and provenance; propose a task or configuration; apply a constraint check; handle uncertainty or a fault conservatively; and preserve evidence for later review. That pattern is valuable as a reusable engineering discipline, but it is too general to be a defensible patent claim by itself.

Layer	Where it already appears	Opportunity / boundary
Vehicle/platform layer	AEROSYNC, AQUAVANTIS-G, SAGE-APC, UCAS	Keep each domain-specific chassis, sensor, control, and physical embodiment in its own package.
Runtime decision layer	AEROSYNC task allocator; AQUAVANTIS-G mission-feasibility evaluator; UCAS flight planner and safety gate	Avoid making the next concept another universal autonomous planner or motion controller.
Configuration/readiness layer	SAGE-APC configuration ledger; UCAS authenticated model packages and model lifecycle	A cross-platform change-impact mechanism may complement these if it is distinct from identity/authentication and ordinary status ledgers.
Validation/evidence layer	All four packages include safety, validation, logging, or test plans	The clearest portfolio-level gap is a shared requirement/assumption/configuration/test dependency graph and targeted evidence refresh.
Release/governance layer	Human oversight and approval exist in several packages	Make TRACEGATE a pre-deployment/change-review tool, not a tactical command layer or a replacement for certified safety controls.

4. Source integrity and readiness findings

- All four packages are drafting or concept-development materials. Their own disclaimers state that they are not filed applications and do not establish novelty, freedom to operate, safety certification, or engineering performance.
- Inventor names, applicant/assignee, priority information, public-disclosure history, and government/contract funding status remain incomplete or require verification in the supplied materials.

- AEROSYNC's filing-readiness gates are listed as open. SAGE's prior-art memo is explicitly preliminary and flags claim differentiation as necessary. UCAS's revised package corrects drawing/reference consistency but still requires attorney review and fact completion.
- The AQUAVANTIS-G package is materially mislabeled relative to the submarine topic: its title page, README, specification, and validation appendix repeatedly describe a UGV. Resolve whether there is a separate submarine concept before treating AQUAVANTIS-G as an underwater system.
- Several high-level building blocks recur across packages: safety supervisor, uncertainty handling, health-aware adaptation, configuration/status ledgers, communications resilience, and evidence logging. Create a claim-element crosswalk to avoid duplicative or conflicting inventions and to make ownership/inventorship clear.

Maturity rating Use these sources as concept inputs and disclosure prompts. Do not represent them as engineering-validated designs, complete patent searches, filed applications, or granted rights.

5. Candidate next-system opportunities

The attached spreadsheet contains the editable weighted matrix. Scores are initial judgment calls on a 1-5 scale, not empirical market research. The weighting emphasizes strategic fit, distinctiveness hypothesis, reusability, feasibility, and evidence readiness.

Rank (initial)	Candidate	Why it matters	Primary risk	Recommendation
1	TRACEGATE: configuration-linked autonomy verification and change-impact gate	Turns the existing safety/validation appendices into a cross-project tool; can be prototyped before building new vehicle hardware.	Crowded neighborhood around robot assurance, authentication, digital twins, and software test selection.	Advance to disclosure workshop + narrow prior-art search.
2	Configuration/ maintenance passport for modular autonomous systems	Builds on SAGE-APC's configuration-and-condition ledger and common fleet configuration data.	Potentially overlaps existing robot verification/authentication and configuration-status systems; SAGE already owns the nearest project-level version.	Treat as a TRACEGATE input, not a standalone next patent unless a distinct mechanism emerges.
3	Cross-platform data-quality and sensor-disagreement evidence service	Standardizes sensor freshness, provenance, uncertainty, and data-quality labels across vehicle types.	Sensor fusion and data provenance are established areas; the technical delta must be concrete.	Keep as a submodule or second-wave concept.
4	Scenario-driven digital-twin and regression-validation bench	Could speed repeatable validation and fault injection across platform updates.	Autonomous-vehicle digital twin calibration is an active patent area; broad scope is risky.	Build only if TRACEGATE needs it and the specific mechanism is distinguishable.
5	Energy-aware degraded-mode recovery manager	Addresses safe recovery when communications, energy, or a component degrades.	High overlap with the UCAS package and published health-aware UAS contingency work.	Do not prioritize as a separate family without a claim chart.
6	New domain-specific vehicle subsystem (e.g., all-weather UGV perception)	Can create physical differentiation and measurable performance improvements.	Needs hardware, operating-domain definition, test access, and greater engineering investment.	Defer until the target platform identity and available test resources are confirmed.

6. Recommended development hypothesis: TRACEGATE

Working name: TRACEGATE - Configuration-Linked Autonomy Verification and Change-Impact Gate. The intended product is a platform-neutral engineering and release-assurance system for authorized civilian and safety-focused robotic applications. Its core function is to answer: 'Given this exact configuration and proposed change, which safety and reliability claims no longer have current evidence, which tests must be run, and what evidence is required before this configuration is approved for the defined use?'

Proposed core mechanism

1. Create a versioned configuration snapshot: hardware revisions, sensors and calibration, payloads/modules, software and model identifiers, operating envelope, approved limits, and maintenance/inspection state.
2. Maintain a dependency graph linking each configuration item to requirements, assumptions, hazards, test scenarios, acceptance criteria, and evidence records.
3. When a change event occurs, propagate its impact through the graph and identify which claims/tests are affected, which evidence remains valid, and which evidence is stale, missing, contradictory, or outside its approved domain.
4. Select a minimal but justified test set using dependency, safety criticality, change type, uncertainty, and historical failure relevance; preserve an explanation for why each test was selected or omitted.
5. Collect reproducible test evidence with source/configuration hashes, software/model version, environment, pass/fail result, uncertainty, and reviewer disposition.
6. Issue a readiness result for a defined configuration and use envelope: READY, READY WITH RESTRICTIONS, BLOCKED/UNKNOWN, or REVIEW REQUIRED. A change or expired/contradicted evidence can invalidate the prior readiness result.

The distinguishing hypothesis to investigate is not 'a safety score' or 'a digital twin.' It is the specific, deterministic dependency and evidence-refresh workflow that transforms a typed configuration change into auditable, targeted proof obligations and a bounded readiness decision. This hypothesis must be checked against actual claims and other published technical material.

What TRACEGATE should not do

- It should not generate or command routes, task allocations, flight trajectories, or tactical behavior.
- It should not replace an onboard independent safety supervisor, human operator, regulator, maintenance inspection, or formal certification.
- It should not equate a passing test suite with absolute safety; it reports the configuration, evidence, assumptions, test limits, and unresolved uncertainties.
- The initial use cases should be non-weaponized inspection, mapping, environmental monitoring, research, industrial mobile robotics, and emergency-response assessment.

7. Development plan and objective gates

Gate	Work product	Exit criteria	Stop / redirect trigger
G0 - identity and ownership	Source-of-truth register for all four packages; platform identity resolved; inventor/disclosure/funding questionnaire.	Each concept has a defined version, intended use, known contributors, rights status, and disclosure history.	Unresolved ownership, conflicting source identity, or undocumented critical concepts.
G1 - problem specification	TRACEGATE one-page problem statement, data model,	At least one measurable workflow improvement is defined and	Concept cannot be distinguished from ordinary requirements

Gate	Work product	Exit criteria	Stop / redirect trigger
	dependency-graph diagram, 3 worked examples, baseline workflow.	compared with a simple full-regression baseline.	management/configuration tracking.
G2 - prototype	Small software prototype with versioned config, change propagation, test selection, evidence ledger, and readiness states.	Reproducible trace from one sample change to impacted tests, evidence status, and a explainable result.	False-clear cases, stale evidence, or unexplained omissions in safety-critical test selection.
G3 - validation	Scenario suite; synthetic or benign test records; fault injection for missing/stale/conflicting evidence; performance report.	Measured detection of impacted evidence, repeatable test selection, traceability, and rollback/revocation behavior.	Results do not improve on a baseline or depend on undocumented manual exceptions.
G4 - patent search	Claim-focused search and chart against nearest patents and non-patent literature; invention disclosure.	Every candidate claim maps to a specific technical mechanism and supporting embodiment; counsel sees a plausible, narrowed search target.	Closest art covers the core sequence without a defensible distinction.
G5 - filing decision	Counsel-reviewed specification/drawings/claims and business decision memo.	Inventorship, assignment, priority, disclosure timing, government rights, and filing jurisdiction reviewed.	Insufficient written-description support or no meaningful technical differentiation.

Suggested first validation examples: (1) swapping a sensor or changing its calibration; (2) installing or removing a modular payload; (3) upgrading an autonomy model or safety-critical library. For each, TRACEGATE should identify affected assumptions and tests, retain unaffected evidence only when its validity conditions remain satisfied, and prevent a missing/expired/contradictory item from yielding an unqualified READY result.

8. Patent and evidence work to do before drafting claims

- Create a feature crosswalk across AEROSYNC, AQUAVANTIS-G, SAGE-APC, and UCAS; mark each feature as shared foundation, platform-specific implementation, or proposed new mechanism.
- Search claims, families, citations, prosecution histories, and non-patent literature for requirement/evidence graphs, change-impact analysis, regression-test selection, safety-case automation, configuration verification, robot authentication, and autonomy release gating.
- Build a closest-reference chart: claim element, source paragraph/figure, closest reference passage/claim, remaining distinction, and technical effect.
- Document specific implementation detail: graph schema, typed change events, dependency traversal, evidence freshness and applicability rules, test-selection logic, readiness-state transitions, conflict handling, and concrete example outputs.
- Record prototype results and baselines. A general goal such as 'improves safety' or 'reduces testing' is not a substitute for the technical mechanism and evidence.
- Have patent counsel assess inventorship, ownership/assignment, public disclosure, priority, government or contractor rights, and provisional versus nonprovisional strategy.
- Keep freedom-to-operate separate from patentability. A patent on a new improvement does not automatically grant a right to practice every underlying component.

9. Preliminary public landscape watchlist

A limited web screen on 19 October 2026 surfaced the following records as search leads. They are not a complete search, claim chart, or legal conclusion. The status, claim scope, related family, and legal events must be confirmed from official records and with counsel.

Public record	Why it matters to the recommended concept	URL
US12475470B1 - Verification and authentication of robots (published 18 Nov 2025)	Nearby for robot verification, configuration/software/compliance information, and restricted operation when component identity does not match. TRACEGATE must differentiate from identity/authorization and digital-passport functions.	https://patents.google.com/patent/US12475470B1/en
US20240095354A1 - Assurance model for an autonomous robotic system (published 21 Mar 2024)	Nearby for multi-layer assurance assessment covering hardware, software, AI, and suppliers. Avoid relying on a generic assurance score as the main distinction.	https://patents.google.com/patent/US20240095354A1/en
US20230080379A1 - Digital twin for an autonomous vehicle (published 16 Mar 2023)	Nearby for calibrating vehicle digital twins against physical-test data and using them to tune controllers. TRACEGATE should not rest on 'a digital twin' alone.	https://patents.google.com/patent/US20230080379A1/en
US20260127062A1 - Health-aware dynamic contingency management for autonomous UAS	Nearby for coupled health state, communications faults, energy margins, and contingency selection. Supports deprioritizing a separate energy/recovery patent absent strong differentiation.	https://patents.google.com/patent/US20260127062A1/en
AEROSYNC package's cited UAV task-allocation references	The supplied checklist cites US10203701B2 and CN121386898A as search leads around multi-UAV task assignment and conflict resolution.	https://patents.google.com/patent/US10203701B2/en

10. Decisions required from the owner

- Confirm whether AQUAVANTIS-G is intended to be a UGV, an autonomous submarine, or two separate concepts. Do not blend embodiments across vehicle classes without deliberate disclosure.
- Confirm the intended product buyer and operating setting for TRACEGATE: robotics manufacturer, testing lab, integrator, fleet operator, or maintenance/quality organization.
- Identify which current concepts have actual sketches, code, prototype results, test logs, or specific decision logic beyond the generated draft documents.
- List each real contributor and the features they conceived; identify employment, contractor, university, grant, or government obligations.
- List all public disclosures, demonstrations, online uploads, sales/offer-for-sale events, and prior filings by date.
- Choose the first benign proof-of-concept embodiment (recommended: a civilian quadrotor or UGV configuration update) and define baseline test effort.

11. Source register and limits

Package	Primary source files reviewed	How used
AEROSYNC	00_AEROSYNC_Complete_Patent_Development	Platform scope, autonomy and task-allocation

Package	Primary source files reviewed	How used
	_Package.pdf; 01_Patent_Specification_Draft.docx; 02_Engineering_Safety_Validation_Appendix.docx; 07_Preliminary_Prior_Art_and_Filing_Checklist.pdf	features, safety and validation plan, prior-art warnings.
AQUAVANTIS-G	00_AQUAVANTIS_G_Complete_Patent_Development_Package.pdf; 01_Patent_Specification_Draft.docx; 02_Engineering_Safety_Validation_Appendix.docx; 04_Preliminary_Prior_Art_and_Filing_Checklist.pdf; README_and_Disclaimer.txt	UGV content, mission feasibility, supervisor, energy/payload handling, and source identity discrepancy.
SAGE-APC	SAGE_APC_Patent_Specification_Draft.docx; SAGE_APC_Engineering_Development_Annex.docx; SAGE_APC_Preliminary_Prior_Art_and_Filing_Memo.docx	Configuration-and-condition ledger, crew-safety integration, and adjacent modular vehicle/condition sensing art.
UCAS revised	01_US_Utility_Specification_Claims_Abstract_REVISED.docx; 06_Invention_Disclosure_Claim_Strategy_REVISED.docx; 08_README_Error_Resolution_Log_REVISED.docx; 04_IDS_Draft_Preliminary_Prior_Art_REVISED.docx	Confidence-gated aircraft autonomy, model/evidence lifecycle, mission state reconciliation, and overlap warnings.

Limitations: The source materials were treated as user-supplied concept drafts and inspected for their stated technical scope, gaps, and warnings. No prototype was tested. The public-record screen is preliminary and limited. No patentability, infringement, freedom-to-operate, regulatory, or legal opinion is provided.